

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Design a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern, responsive webpage using **HTML5 and CSS3** with a navigation bar, hero section, about section, services section, contact form, and footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
 - To design a webpage using HTML5 semantic elements.
 - To apply CSS3 styling to HTML elements.
 - To create a responsive navigation bar.
 - To design an attractive hero section using CSS.
 - To create service cards using CSS Flexbox.
 - To design and style a contact form.
 - To implement hover effects and smooth scrolling.
 - To make the webpage responsive using CSS media queries.
 - To understand the separation of HTML structure and CSS presentation.
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4. THEORY

HTML5 (HyperText Markup Language 5) is used to define the structure and content of a webpage. It provides elements such as `<header>`, `<nav>`, `<section>`, `<form>`, and `<footer>` for organizing webpage content.

CSS3 (Cascading Style Sheets 3) is used to control the presentation and appearance of HTML elements. CSS can be used to define colors, fonts, spacing, layouts, backgrounds, borders, animations, and responsive behavior.

In this experiment, HTML5 is used to create the structure of a website called **MyWebsite**, while an external CSS file named `styles.css` is used for styling.

The webpage consists of:

- **Header:** Contains the website logo and navigation links.
- **Hero Section:** Contains the main welcome message and Learn More button.
- **About Section:** Provides information about the website.
- **Services Section:** Displays three service cards.
- **Contact Section:** Contains a form for entering user information.
- **Footer:** Displays copyright information.

CSS3 features such as **Flexbox, gradients, transitions, hover effects, sticky positioning, box shadows, and media queries** are used to improve the appearance and responsiveness of the webpage.

The basic working flow is:

HTML Structure → CSS Styling → Browser Rendering → Responsive Webpage

5. ALGORITHM / PROCEDURE

1. Create a new project folder for the webpage.
2. Create an HTML file named `index.html`.
3. Create an external CSS file named `styles.css`.
4. Define the HTML5 document structure using `<!DOCTYPE html>`.
5. Add the webpage title and link the external CSS file.
6. Create a header containing the website logo and navigation links.
7. Create a hero section with a heading, description, and button.
8. Create an About section containing information about the website.
9. Create a Services section containing three service cards.
10. Create a Contact section containing a form with name, email, and message fields.
11. Create a footer containing copyright information.
12. Use CSS to style the webpage elements.
13. Apply Flexbox to arrange navigation elements and service cards.
14. Apply colors, spacing, typography, gradients, borders, and shadows using CSS.
15. Add hover effects to navigation links, buttons, and service cards.
16. Add smooth scrolling using the `scroll-behavior` property.
17. Use `position: sticky` to keep the header visible while scrolling.

18. Implement responsive design using the `@media` rule.
 19. Save both HTML and CSS files in the same project folder.
 20. Open `index.html` in a web browser and verify the complete webpage output.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Webpage</title>
  <link rel="stylesheet" href="styles.css">
</head>

<body>

  <!-- Header -->
  <header>
    <div class="logo">MyWebsite</div>

    <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Hero Section -->
  <section class="hero" id="home">
    <div class="hero-content">
      <h1>Welcome to My Website</h1>
      <p>
        A simple and modern webpage designed using HTML5 and CSS3.
      </p>
      <a href="#about" class="btn">Learn More</a>
    </div>
  </section>

  <!-- About Section -->
  <section class="about" id="about">
    <h2>About Us</h2>
```

```
<p>
    We create clean, responsive and user-friendly websites
    using modern web technologies.
```

```
</p>
</section>
```

```
<!-- Services Section -->
<section class="services" id="services">
    <h2>Our Services</h2>
```

```
    <div class="service-container">

        <div class="card">
            <h3>Web Design</h3>
            <p>
                Attractive and responsive website designs for
                all types of devices.
```

```
            </p>
        </div>
```

```
        <div class="card">
            <h3>Development</h3>
            <p>
                Modern websites developed using HTML5, CSS3
                and JavaScript.
```

```
            </p>
        </div>
```

```
        <div class="card">
            <h3>Support</h3>
            <p>
                Reliable maintenance and support for your
                website.
```

```
            </p>
        </div>
```

```
    </div>
</section>
```

```
<!-- Contact Section -->
<section class="contact" id="contact">
    <h2>Contact Us</h2>
```

```
    <form>
        <input type="text" placeholder="Your Name" required>
        <input type="email" placeholder="Your Email" required>
```

```
        <textarea placeholder="Your Message" rows="5"></textarea>
        <button type="submit">Send Message</button>
    </form>
</section>
```

```
<!-- Footer -->
<footer>
    <p>&copy; 2026 MyWebsite. All Rights Reserved.</p>
</footer>
```

```
</body>
</html>
```

```
styles.css * {    margin: 0;
padding: 0;    box-sizing:
border-box;    scroll-
behavior: smooth;
}
```

```
body {    font-family: Arial,
sans-serif;    line-height: 1.6;
color: #333;    background-
color: #f5f5f5;
}
```

```
/* Header */ header {
background-color: #222; color:
white; display: flex; justify-
content: space-between; align-
items: center; padding: 20px
8%; position: sticky; top: 0;
z-index: 1000;
}
```

```
.logo {    font-size:
26px; font-weight:
bold;
}
```

```
nav {    display:
flex;    gap:
25px;
}
```

```
nav a {    color: white;
text-decoration: none;
```

```
font-size: 16px;  
transition: 0.3s;  
}
```

```
nav a:hover {  
color: #00adb5;  
}
```

```
/* Hero Section */  
.hero { height: 90vh;  
display: flex; justify-  
content: center; align-items:  
center; text-align: center;  
background: linear-gradient(  
    135deg,  
    #00adb5,  
    #393e46  
);  
color: white;  
padding: 20px;  
}
```

```
.hero-content { max-  
width: 700px;  
}
```

```
.hero h1 { font-size:  
50px; margin-bottom:  
20px;  
}
```

```
.hero p { font-size:  
20px; margin-bottom:  
30px;  
}
```

```
.btn { display: inline-  
block; background-color:  
white; color: #222;  
padding: 12px 25px;  
border-radius: 5px; text-  
decoration: none; font-  
weight: bold; transition:  
0.3s;  
}
```

```
.btn:hover { background-color: #222; color: white;
}
```

```
/* Common Section Styles */
section { padding: 70px 8%; text-align: center;
}
```

```
section h2 { font-size: 35px; margin-bottom: 20px;
}
```

```
/* About */ .about { background-color: white;
}
```

```
.about p { max-width: 700px; margin: auto; font-size: 18px;
}
```

```
/* Services */ .services { background-color: #f5f5f5;
}
```

```
.service-container { display: flex; justify-content: center; gap: 25px; margin-top: 35px; flex-wrap: wrap;
}
```

```
.card { background-color: white; width: 300px; padding: 30px; border-radius: 10px; box-shadow: 0 5px 15px rgba(0, 0, 0, 0.1); transition: 0.3s;
}
```

```
.card:hover { transform: translateY(-8px);
}
```

```
.card h3 { color:
#00adb5; margin-
bottom: 15px; font-
size: 23px;
}

/* Contact */
.contact { background-
color: white;
}

form { max-width:
600px; margin: 30px
auto 0; display: flex;
flex-direction: column;
gap: 15px; }

input, textarea {
width: 100%; padding:
14px; border: 1px
solid #ccc; border-
radius: 5px; font-size:
16px; font-family:
inherit;
}

textarea {
resize: vertical;
}

button { padding: 14px;
border: none; border-radius:
5px; background-color:
#00adb5; color: white;
font-size: 17px; cursor:
pointer; transition: 0.3s;
}

button:hover { background-
color: #008b91;
}

/* Footer */ footer {
background-color: #222;
color: white; text-align:
center; padding: 20px;
}
```



```

/* Responsive Design */ @media (max-width: 768px) {

    header {
        flex-direction: column;
        gap: 15px;
    }

    nav {
        gap: 15px;
        flex-wrap: wrap;
        justify-content: center;
    }

    .hero h1 {
        font-size: 36px;
    }

    .hero p {
        font-size: 17px;
    }

    section {
        padding: 50px 5%;
    }

    .service-container {
        flex-direction: column;
        align-items: center;
    }

    .card {
        width: 100%;
        max-width: 350px;
    }
}

```

7. CODE EXPLANATION

Code	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.
<meta charset="UTF-8">	Defines the character encoding as UTF-8.
<meta name="viewport"...>	Makes the webpage responsive on different screen sizes.

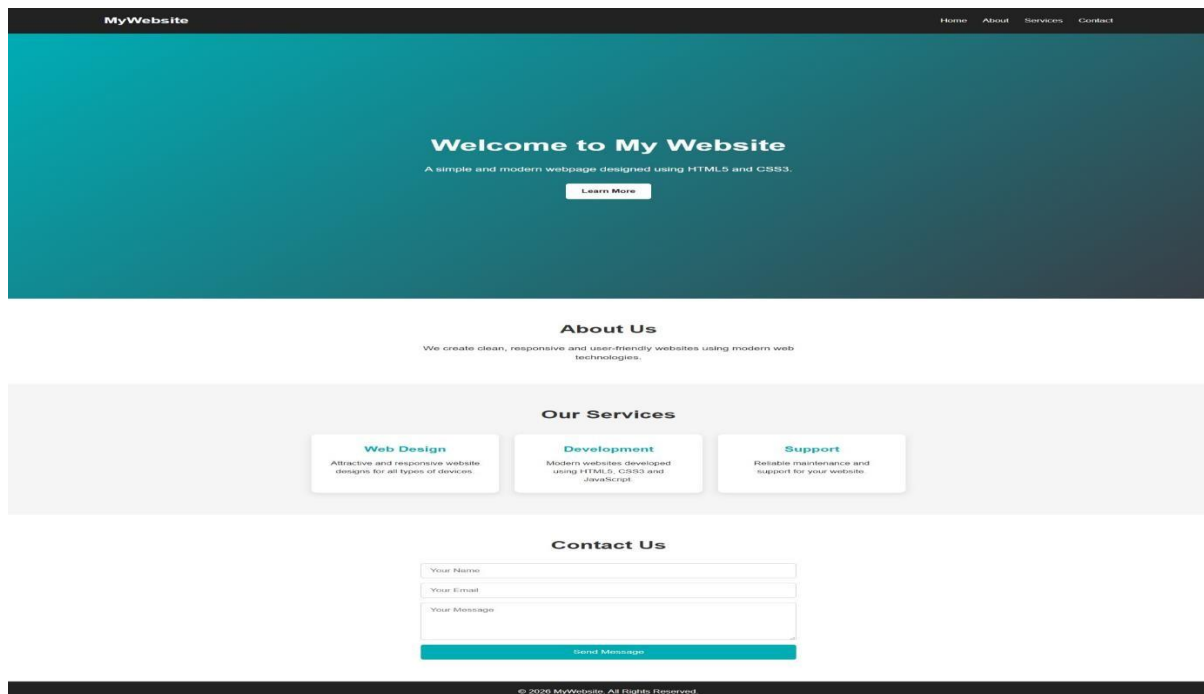
<title>	Specifies the title displayed in the browser tab.
<link rel="stylesheet"...>	Connects the external CSS file with the HTML document.
<header>	Defines the header section of the webpage.
<nav>	Contains the main navigation links.
<section>	Defines individual sections of the webpage.
id	Provides a unique identifier for internal navigation.
<div>	Groups related HTML elements together.
<a>	Creates hyperlinks and navigation links.
<form>	Creates the contact form.
<input>	Accepts user input such as name and email.
<textarea>	Provides a multi-line input area for messages.
<button>	Creates the form submission button.
<footer>	Defines the footer section.
display: flex	Creates a flexible layout for arranging elements.
justify-content	Controls the horizontal alignment of flex items.
align-items	Controls the vertical alignment of flex items.
gap	Adds spacing between flex items.
background-color	Sets the background color of an element.
background: lineargradient()	Creates a gradient background for the hero section.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow effect to service cards.
transition	Creates smooth visual transitions.
:hover	Applies styles when the mouse pointer moves over an element.
position: sticky	Keeps the header fixed within its scrolling position.
scroll-behavior: smooth	Provides smooth scrolling between page sections.
@media	Defines responsive styles for different screen sizes.
flex-wrap	Allows flex items to move to the next line when required.

8. TEST CASES AND EXECUTION DATA

The webpage was executed in a web browser and tested for navigation, layout, styling, form controls, hover effects, and responsive behavior.

Test Case	TC-01
Test / Input	Load the webpage in the browser
Expected Result	Header, navigation bar, hero section, About Us section, Our Services section, Contact Us form, and footer should be displayed correctly.
Actual Result	All webpage sections were displayed correctly.
Status	Pass

TC-01 Output:

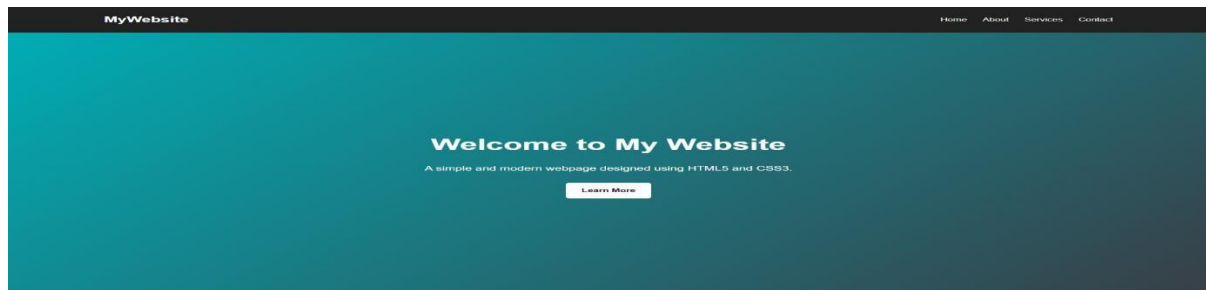


Test Case	TC-02
Test / Input	Check the navigation bar containing Home, About, Services, and Contact links
Expected Result	All navigation links should be visible and should navigate to their respective sections of the webpage.
Actual Result	Home, About, Services, and Contact navigation links were displayed correctly.
Status	Pass

TC-02 Output:

Test Case	TC-03
Test / Input	Check the hero section and "Learn More" button
Expected Result	The hero section should display "Welcome to My Website", the description, and the "Learn More" button correctly.
Actual Result	The hero heading, description, and "Learn More" button were displayed correctly.
Status	Pass

TC-03 Output:



Test Case	TC-04
Test / Input	Check the About Us section
Expected Result	The About Us heading and its description should be displayed correctly.
Actual Result	The About Us section and description were displayed correctly.
Status	Pass

TC-04 Output:



Test Case	TC-05
Test / Input	Check the Our Services section
Expected Result	Three service cards for Web Design, Development, and Support should be displayed with their descriptions.
Actual Result	All three service cards and their descriptions were displayed correctly.
Status	Pass

TC-05 Output:



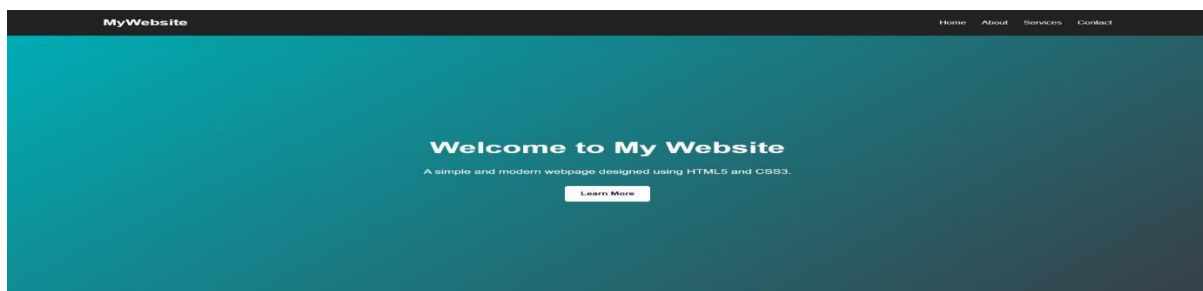
Test Case	TC-06
Test / Input	Enter sample data into the Contact Us form
Expected Result	The user should be able to enter a name, email address, and message in the Contact Us form.
Actual Result	Sample data was entered successfully into all contact form fields.
Status	Pass

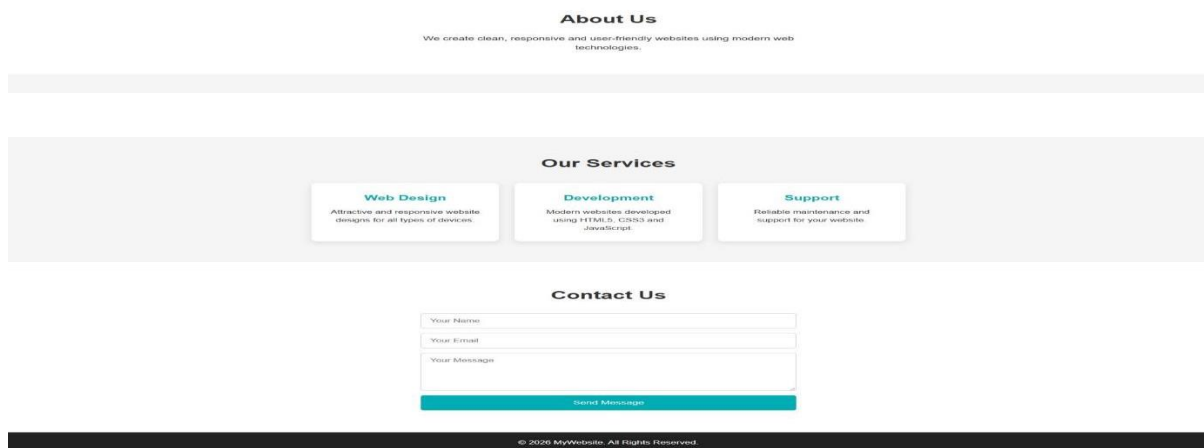
TC-06 Output:

Contact Us

Test Case	TC-07
Test / Input	Click the Home, About, Services, and Contact navigation links
Expected Result	Each navigation link should move the user to its corresponding section of the webpage.
Actual Result	The navigation links directed to their respective webpage sections correctly.
Status	Pass

TC-07 Output:





Test Case	TC-08
Test / Input	Check the footer section
Expected Result	The footer should be displayed at the bottom of the webpage with the copyright information.
Actual Result	The footer and copyright information were displayed correctly.
Status	Pass

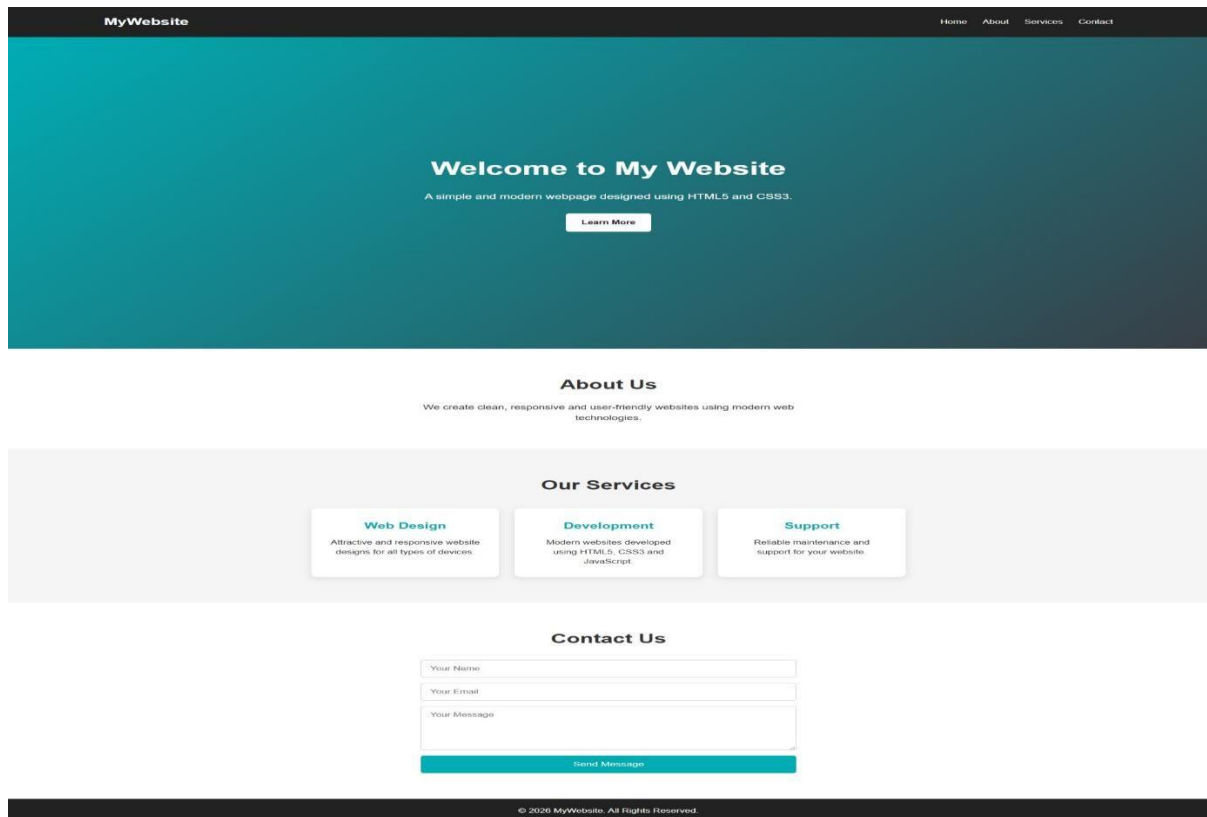
TC-08 Output:



9. OUTPUT

Output: Complete Responsive Webpage

The following screenshot shows the complete execution of the **HTML5 and CSS3 webpage** in the web browser. It contains the header and navigation bar, hero section, About section, Services section, Contact form, and footer.



10. RESULT

Thus, a modern and responsive webpage was successfully designed and developed using **HTML5 and CSS3**. The webpage contains a navigation bar, hero section, About section, service cards, contact form, and footer. CSS3 features such as Flexbox, gradients, transitions, hover effects, sticky positioning, box shadows, and media queries were successfully implemented. The webpage was executed and verified successfully in a web browser.